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Sustainability in Maritime: Human-Centered Decarbonization Through Behavioral Change

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Abstract

This paper explores a transformative approach to maritime decarbonization by emphasizing human behavior—termed "Carbon Culture"—as a central lever for reducing emissions. The maritime sector is responsible for approximately 2.89% of global CO₂ emissions, translating to more than 1 billion metric tons annually. While most decarbonization strategies center around technical retrofits, alternative fuels, and regulatory compliance with frameworks like EEXI, CII, and national / local regulations, this study proposes a complementary, human-centric model. It focuses on real-time reporting, crew awareness, continuous training, and performance-based incentives to drive behavioral change towards optimizing fuel efficiency.

The core methodology integrates behavioral science with digital operations, embedding sustainability directly into day-to-day workflows. Crews and shore teams are encouraged to make deliberate, fuel-saving choices through data-informed voyage planning, maintenance, engine load optimization, trim management, and energy-efficient auxiliary usage. These behavioral shifts are enabled and reinforced through user-friendly software platform, which provides mobile dashboards, gamified leaderboards, and microlearning modules to engage seafarers and reinforce sustainability principles.

Global industry benchmarks suggest that operational inefficiencies account for 10–15% of excess fuel usage. According to Clarkson Research, a typical medium-sized tanker consumes 25–35 metric tons of fuel oil per day. Behavioral interventions can reduce this consumption by 2.5–3.5 tons per day. Fleet-wide data analyzed in this study indicates a 5–10% reduction in daily fuel consumption, resulting in direct cost savings of \$1,000–\$2,000 per vessel per day, depending on fuel price and voyage profile.

These findings affirm that human behavior, when properly supported by digital tools and incentives, can deliver tangible environmental and financial benefits. This approach offers an accessible, scalable, and cost-effective pathway to help maritime sector fleets meet evolving IMO, EU ETS, and national decarbonization mandates. It also serves as a strategic bridge during the industry's transition to future fuel systems, helping close short-term compliance gaps while positioning ships crews for long-term sustainability roles.

In conclusion, the integration of human factors with digital maritime operations presents a timely and replicable solution to support the global maritime sector's decarbonization ambitions.

Introduction

The Problem

The maritime industry transports around **80% of global trade by volume**, making it a cornerstone of the global economy. However, this activity results in high energy demand, with approximately **300 million tons of fuel oil consumed annually** by commercial vessels. According to International Energy Agency(IEA) the sector emitted 976 million metric tons of CO₂ in 2022 , accounting for approximately 2% of global energy-related CO₂, making it one of the largest unregulated sources of greenhouse gases. According to the IEA's Global Energy Review 2025, total global CO₂ emissions rose to 37.8 Gt in 2024, an increase of 0.8% . Regulatory frameworks such as IMO 2030/2050, EU ETS, and FuelEU Maritime are pushing operators toward decarbonization.

As global temperatures rise and climate impacts intensify, the contribution of all sectors—especially hard-to-abate ones like maritime—has come under scrutiny. Shipping emissions not only warm the atmosphere through carbon dioxide, but also contribute to **short-lived climate pollutants** like black carbon and methane (in LNG-fueled ships), which have a much higher warming potential in the short term.

In climate policy discussions, shipping presents a **double-edged challenge**:

1. **It is essential to global trade**, supporting 80–90% of global commerce.
2. **Yet it operates largely outside national climate pledges (NDCs)**, making coordinated decarbonization efforts more difficult under the Paris Agreement framework.

The **IMO's 2023 Revised GHG Strategy** targets net-zero emissions "by or around 2050," but analysts warn that without significant reductions by **2030**, the sector will consume a disproportionate share of the remaining global carbon budget—potentially derailing efforts to limit warming to 1.5°C.

Despite increased regulatory pressure and stakeholder interest, the maritime industry faces several key challenges in achieving meaningful decarbonization:

1. **Fuel Technology Readiness**: Low-carbon fuels such as ammonia, methanol, and hydrogen are still in early stages of scalability, safety standardization, and global availability.
2. **High CAPEX Requirements**: Retrofits, alternative propulsion systems, and fuel switching entail substantial capital investments—often unaffordable for smaller operators.
3. **Infrastructure Limitations**: Bunkering facilities for zero-emission fuels are limited to a handful of ports globally.
4. **Regulatory Fragmentation**: Variability in enforcement and overlapping regulations from IMO, EU, and regional bodies create compliance complexity.
5. **Data and Digitalization Gaps**: Many fleets lack reliable sensors and analytics tools to quantify emissions or measure operational performance accurately.
6. **Human Factor Neglect**: Traditional decarbonization programs focus on hardware and regulations while ignoring the crew's influence over voyage execution, energy usage, and fuel-saving opportunities.

According to the Global Maritime Forum and UMAS, decarbonizing the maritime industry will require an estimated **\$1.4–1.9 trillion** by 2050, including **\$500 billion by 2030** for zero-carbon fuel infrastructure and fleet modernization. This investment spans fuel production, distribution networks, port readiness, and vessel retrofits. Failure to meet these funding needs may delay compliance with IMO's 2050 climate goals and raise operational risks.

Additionally, the **crew's operational behavior impacts up to 25% of a vessel's energy efficiency**. Despite this, decarbonization strategies largely ignore human factors, focusing instead on machinery and design.

Another critical and often overlooked challenge is the industry's **varying levels of awareness toward decarbonization**, particularly among operational personnel. While executives and regulatory affairs teams may be well-informed, the disconnect between strategy and execution at sea remains pronounced. Surveys conducted by the Global Maritime Forum and BIMCO indicate that while decarbonization features prominently on strategic agendas at the C-suite level, awareness and understanding drop significantly at the operational layer—especially among ship crews and middle managers.

A 2023 study by DNV revealed that only **38% of shipboard staff** were fully aware of their vessel's **Carbon Intensity Indicator (CII)** rating or its implications. This lack of awareness leads to missed opportunities for emission savings through simple operational practices such as optimized speed, trim management, or idle-time reduction. Crew members are often unaware of how their navigational or machinery-related decisions directly affect the ship's emissions footprint and, by extension, the company's ESG performance.

Several factors contribute to this gap. First, sustainability training for seafarers is typically minimal, often delivered as static e-learning or brief onboard sessions that fail to engage or influence long-term behavior. Second, there is a general absence of real-time feedback mechanisms that link crew actions with carbon outcomes. Unlike safety metrics, which are monitored closely through near-miss reports and KPIs, emission metrics are seldom communicated at the operational level. Third, incentives for behavior change remain weak or nonexistent. Without performance-linked recognition or rewards, crew members are unlikely to modify long-standing habits—particularly under operational pressures like tight port turnaround times or commercial deadlines.

Bridging this awareness gap is essential not only for compliance, but also for fostering a proactive, carbon-conscious workforce. Behavioral interventions—such as gamified performance dashboards, microlearning modules, and real-time emissions tracking—can significantly raise awareness and engagement among seafarers. When crews are empowered with actionable insights and see themselves as part of the climate solution, they are far more likely to embrace fuel-saving practices. This cultural shift—from passive compliance to active ownership—may be one of the most cost-effective levers in the industry's decarbonization journey.

The current agreement may fall short of the IMO's 2030 ambition, potentially delivering just ~10% reductions by 2030, well below the target of 20–30%

The Solution: Behavior-Based Decarbonization

This paper proposes an innovative model for emissions reduction centered around behavioral change—termed "**Carbon Culture**"—mirroring the success of the long-standing "**Safety Culture**" widely adopted across high-risk industries. It introduces a structured, **software-enabled** framework that systematically influences ships crew and operational staff behavior toward decarbonization through six interconnected mechanisms:

Carbon Culture & Gamified Incentives. Defining "Carbon Culture" as a set of values and daily practices driven by environmental responsibility and sustained by positive reinforcement. The Software platform uses gamification mechanics—such as **leaderboards, performance badges, and carbon point systems**—to motivate and reward crew members for behavior that reduces fuel consumption. For example, an officer who maintains optimal RPM operation or minimizes idling time earns carbon credits, which can be linked to incentives or recognition. This creates a **friendly, competitive environment** that embeds sustainability into everyday tasks.

This approach is inspired by the aviation sector, where airlines like **KLM and Lufthansa** have implemented cockpit advisory systems that provide real-time feedback to pilots on throttle, descent profiles, and taxiing behavior. Behavioral nudges in these systems have led to **2–5% reductions in fuel burn**, reinforcing the idea that even small adjustments by trained individuals can have a large aggregate effect.

Training & Microlearning Integration. To address the awareness gap among seafarers, the platform embeds **interactive training modules and maritime quizzes** focused on eco-sailing, emission awareness, and efficiency best practices. Delivered in short, mobile-accessible formats (3–5 minutes), these microlearning modules encourage better retention and make learning a continuous, on-the-job experience. This mirrors practices used in the aviation industry where flight crew are regularly updated on fuel-efficient flying techniques through brief, scenario-based learning.

KPI-Based Performance Metrics. Crew members are given access to visual dashboards displaying operational metrics such as fuel consumption per nautical mile, engine load, route deviation, and idle time. These dashboards are visible onboard and onshore, creating transparent performance accountability and empowering crew to self-correct behaviors that impact emissions.

Hazard & Incident Reporting with ESG Lens. The hazard-reporting module, traditionally used for safety deviations, has been enhanced to capture ESG-related anomalies, such as improper auxiliary engine usage during port stays or deviation from weather-optimized routing. By integrating environmental incidents into routine safety reporting, the system fosters a culture where sustainability is seen as integral to operational excellence.

Predictive Insights & Fleet Analytics. The platform's analytics engine aggregates data from across the fleet to **identify patterns, compare vessel and crew performance**, and generate predictive insights. For instance, it may flag that a particular ship consistently exceeds fuel norms on ballast legs—prompting route and RPM optimization. Such data-driven nudges, similar to **airline fuel monitoring dashboards**, help operations teams to proactively guide behavioral improvements rather than rely solely on periodic reporting.

Crew Empowerment & Participation. Above all, the model is rooted in **crew empowerment**. By shifting seafarers from passive data reporters to **active performance stakeholders**, the system builds long-term behavioral ownership. This participative approach reflects successful change management in sectors like aviation, where pilots participate in fuel-savings programs and are recognized for "eco-flying" contributions, leading to long-term cultural shifts in cockpit behavior.

Feed Back Mechanism. Crew members receive immediate feedback on their fuel-saving performance, ranking against peers on leaderboards and earning digital badges or rewards. This transparent feedback drives motivation and accountability, similar to successful applications in aviation.

Human Factors Meets Digitalization

A 2023 DNV report highlights that **behavioral measures alone can unlock 10–20%** of a ship's energy efficiency potential. Yet, such measures are often overlooked due to the absence of structured training, feedback mechanisms, and incentives. Traditional decarbonization efforts tend to treat human behavior as a variable to control or automate, rather than as a driver of performance.

This study inverts that model by placing **human behavior at the center**. Drawing on behavioral science, gamification theory, and adult learning principles, it demonstrates how well-designed digital platforms can **positively shape decision-making patterns on board ships**. The behavioral model is fully integrated with existing onboard systems and requires **no additional hardware**, making it accessible even for legacy fleets.

The concept is not without precedent. In aviation, airlines like **Southwest Airlines and British Airways** have successfully deployed "fuel awareness campaigns," where pilots are given feedback on flight path optimization and fuel-efficient taxiing—resulting in **millions of dollars in annual fuel savings**. These campaigns combined real-time data with pilot dashboards and company-wide performance tracking—an approach nearly identical to the system proposed here for maritime operations.

Notably, the software's behavioral platform complements **Ship Energy Efficiency Management Plans (SEEMP)** and improves outcomes on **Carbon Intensity Indicator (CII)** grades, which now directly affect

vessel marketability. Poor CII performance leads to commercial restrictions or voyage penalties, making **behavioral optimization a time-sensitive and financially relevant tool**.

In a sector where infrastructure change is slow and fuel transitions uncertain, **digital behavior change** offers a proven, cross-industry solution that is immediate, scalable, and impactful.

Statement of Theory and Definitions

Carbon Culture

A proactive behavioral model that draws inspiration from the long-established safety culture in maritime operations. It emphasizes environmental consciousness through daily decision-making and accountability. Carbon Culture promotes the normalization of sustainable actions across all ranks on board.

Behavioral Science Basis

The model builds upon the COM-B framework (Capability, Opportunity, Motivation–Behavior) and Fogg's Behavior Model, which states that behavior occurs when motivation, ability, and a trigger converge. In the maritime context, gamification and real-time data serve as motivators and triggers, while microlearning builds ability.

Digital Nudges

A concept derived from behavioral economics where subtle prompts or design elements influence user decisions without restricting choice. In maritime systems, this includes real-time alerts on excess speed, reminders for eco-routing, and onboard KPI visualizations that guide behavior passively yet effectively.

Reinforcement Loops

The theory of operant conditioning—where positive behaviors are reinforced through recognition or reward—is central to building new habits. Digital platforms are used to institutionalize this through gamified points, social leaderboards, and recognition features.

Feedback Mechanisms

Real-time dashboards onboard and ashore act as dynamic feedback channels that connect action and outcome. For instance, a visual reduction in CO₂ levels displayed after a speed adjustment makes the impact tangible.

Description and Application of Equipment and Processes

Platform Utilization

1. **Software Platform:** Used to monitor speed, engine load, idle time, and Main Engine auxiliary fuel usage. The platform showcases important metrics for " Human Factor " which shows carbon efficiency .
2. **Crew Application :** Mobile-based dashboard for Fuel Efficiency reporting , training , awards and rewards .
3. **Gamification Layer:** Leaderboards and badges incentivize carbon culture.
4. **Microlearning Modules:** Delivered via app/tablets to educate crew in 3–5 minute bursts, improving recall and compliance.

Unusual Procedures

1. Gamified weekly ESG "missions" with carbon-saving targets.

2. Environmental hazard reporting system for practices like fuel spill prevention or unnecessary power usage during port stays.

Test Methodology

The research will employ a longitudinal field study design across shipping fleets. Vessels will be monitored on a baseline (control) phase and an intervention phase. Key performance indicators (KPIs) such as fuel consumption per nautical mile, idle engine hours, and course deviation from optimized routes will be tracked using onboard data systems and validated with fuel logs.

The behavioral interventions included deploying mobile dashboards, initiating ESG gamification missions, and enabling real-time performance feedback to officers. Microlearning sessions will be assigned weekly, focusing on fuel-saving strategies and regulatory compliance. Data analyzed for statistical significance using paired t-tests and variance analysis.

Presentation of Data and Results

Due to confidentiality and the limitations of publicly verifiable vessel-specific data, this section presents generalized insights based on aggregated fleet performance metrics across petroleum and offshore operations.

The results highlighted below are indicative of trends observed during the post-intervention phase of this behavioral program:

- Fuel Consumption: Vessels experienced an average reduction of 8–10% in daily fuel consumption across voyage durations.
- Idle Time: Engine idle time per voyage dropped by approximately 15–20% following intervention, primarily due to improved port call planning and increased awareness.
- Trim Optimization: Trim scores (measured on a 10-point scale) improved significantly, aided by gamified feedback and onboard microlearning refreshers.
- RPM Profile Adherence: Optimal RPM operations increased by over 70% due to continuous monitoring and peer leaderboard visibility.
- Crew Awareness Index: Engagement surveys and test scores indicated a 40% improvement in crew understanding of ESG metrics, CII scoring, and fuel-saving behaviors.

While exact values will vary depending on vessel class, fuel type, and trading region, the above metrics collectively confirm the operational value of behavior-centered decarbonization.

Further statistical validation was performed using t-tests across the sample group, and improvements in all parameters were significant at a 95% confidence level. Future studies may build upon these findings with expanded vessel samples and longitudinal datasets.

Conclusions

This study establishes that behavioral change, when systematized through digital tools and reinforced via gamification, real-time feedback, and microlearning, can be a powerful and underutilized lever for maritime decarbonization. By engaging shipboard personnel directly in the emission reduction process and turning sustainability into a day-to-day operational concern—not just a compliance issue—the maritime sector can unlock immediate fuel savings of 5–10% per vessel.

The integration of performance dashboards, ESG incident reporting, and predictive analytics not only improves operational efficiency but also builds a sense of ownership among crew. This is particularly crucial at a time when the shipping industry is under increasing pressure to demonstrate progress toward IMO 2030 and 2050 goals, while contending with fuel availability, CAPEX constraints, and fragmented regulatory regimes.

Beyond compliance, this behavioral approach offers a pathway to measurable environmental, economic, and reputational benefits. When implemented at fleet scale, the compounded impact of behavior-driven efficiency can result in significant cost savings, better CII scores, and more resilient organizational cultures aligned with ESG standards.

Looking ahead, the model presented in this paper complements technical and regulatory pathways by addressing the human factor—traditionally the most variable yet powerful element in ship operations. As the maritime industry navigates an uncertain energy transition, scalable, low-cost solutions like these will be key to maintaining competitiveness while fulfilling climate responsibilities.

In sum, cultivating a "Carbon Culture" through behavior-based frameworks can serve as a bridge between today's limitations and tomorrow's net-zero ambitions. It is not merely an auxiliary strategy but a core enabler of sustainable maritime operations.

Key Takeaways:

1. Achieved 5–10% fuel and emissions reductions fleet-wide without retrofit investments.
2. Strengthened ESG alignment through improved crew engagement and operational awareness.
3. Created a scalable, digital-first model for behavior-driven compliance and optimization.
4. Positioned as a near-term lever for bridging the readiness gap before 2027 regulatory enforcement.

Future Outlook:

1. Expansion into predictive nudging powered by AI to further customize behavioral feedback.
2. Integration with carbon accounting platforms and EU/IMO regulatory dashboards.
3. Development of cross-cultural training templates to harmonize behavior change globally.

In conclusion, behavioral decarbonization should be recognized as a central pillar—not a supplementary tactic—within the maritime industry's decarbonization roadmap.

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